Day 5: Network Testing Practical (1 hour): Connect devices to a network and test connectivity (ping, traceroute). Task: Write a report on how to set up and test a basic network.

**1.Connect your laptop and a friend's phone to the same Wi-Fi network, then use the ping command to check if both devices can communicate with each other.<br><br>Hint: On Windows, use 'ping [IP address]' in Command Prompt; on Android, try a ping app or use Termux. **

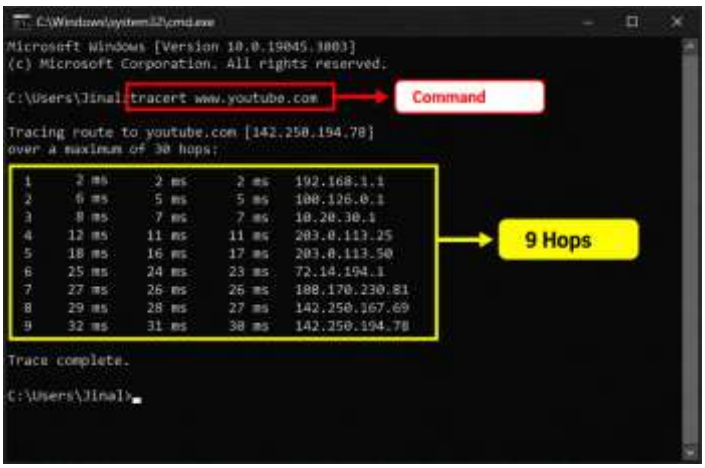
**ANS:**

1. Connect your laptop and your friend's phone to the same Wi-Fi.
2. Find your friend's phone IP address.
3. Open CMD on your laptop.
4. Type ping and the phone's IP address.
5. Press Enter and check the result.
6. If you see Reply, the connection works.
   **CMD :ping 192.168.1.5**

**2.Run the traceroute (tracert on Windows or traceroute on Mac/Linux) command to www.youtube.com from your device and note down the number of hops it takes to reach the destination.**

**ANS:**

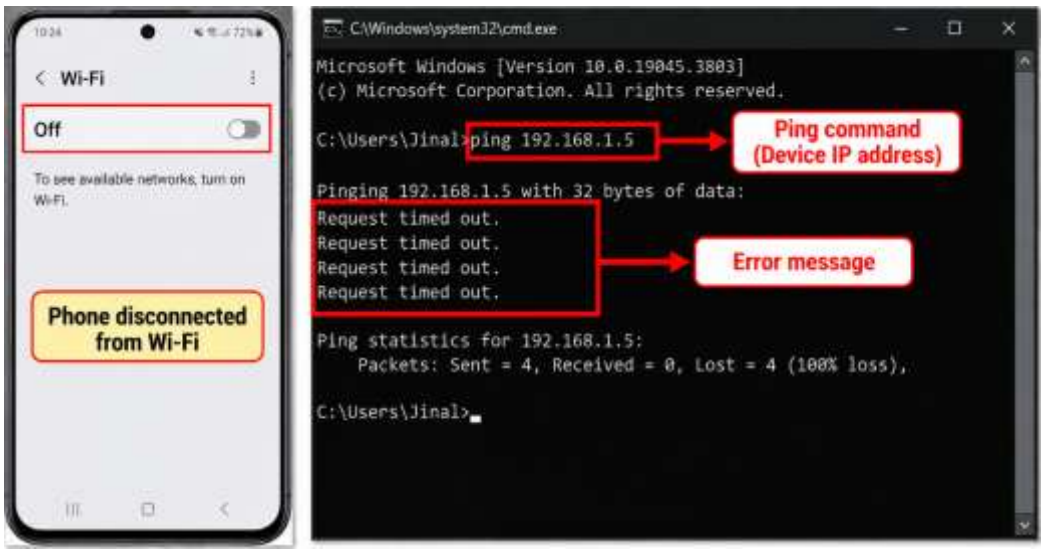
1. Open Command Prompt (CMD) on your laptop.
2. Type this command:
3. tracert www.youtube.com
4. Press Enter.
5. Wait until the command finishes.
6. Count the number of hops shown in the output.



**3.Disconnect one device from the Wi-Fi, then try pinging it again from your laptop and describe the error message you receive.**

**ANS:**

1. Disconnect one device from Wi-Fi.
2. Open CMD on your laptop.
3. Type ping followed by the device's IP address.
4. Press Enter.
5. The message may show Request timed out.
6. This means the device is not replying to the ping.



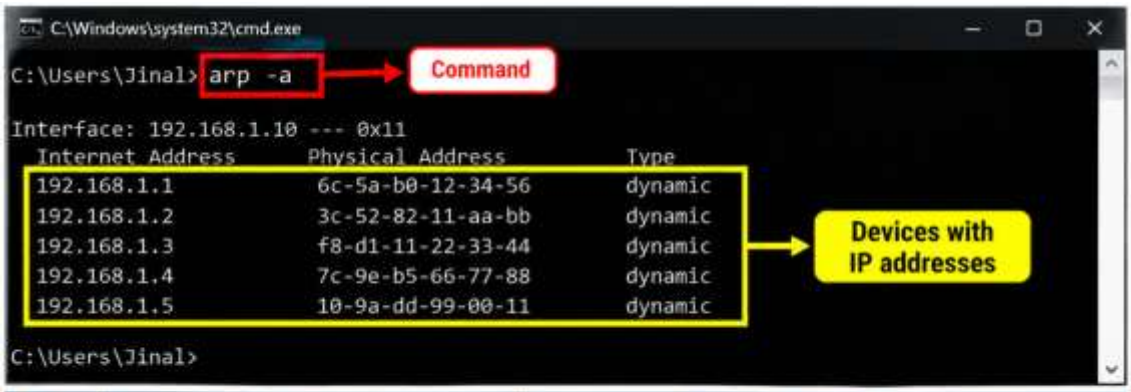

**4.Create a simple table listing the IP addresses of all devices connected to your home Wi-Fi (use 'arp -a' on Windows), and identify which device belongs to which person or gadget.**

**ANS:**

1. Connect your laptop to Wi-Fi.
2. Open CMD.
3. Type arp -a and press Enter.
4. Note the IP addresses shown.

| No. | IP Address (Example) | Device |
|---|---|---|
| 1 | 192.168.1.1 | Wi-Fi Router |

| | | |
|---|---|---|
| 2 | 192.168.1.2 | My Laptop |
| 3 | 192.168.1.3 | My Phone |
| 4 | 192.168.1.4 | Father's Phone |
| 5 | 192.168.1.5 | Smart TV |




| No. | IP Address | Device (Person / Gadget) |
|---|---|---|
| 1 | 192.168.1.1 | Wi-Fi Router |
| 2 | 192.168.1.2 | My Laptop |
| 3 | 192.168.1.3 | My Phone |
| 4 | 192.168.1.4 | Father's Phone |
| 5 | 192.168.1.5 | Smart TV |

## 5.Write a one-page report explaining step-by-step how you set up the network, tested connectivity using ping and traceroute, and what results or issues you observed.<br><br>Hint: Include

## screenshots or command outputs if possible.

## ANS:

### 1. Network Setup

First, I connected my laptop to the Wi-Fi network. I checked the IP address, subnet mask, and default gateway using the ipconfig command in CMD.

### 2. Ping Test

I used the ping command to check the connection between devices. I typed ping followed by the other device's IP address. If I received a reply, the connection was working.

### 3. Traceroute Test

I used the command tracert www.youtube.com in CMD. It showed the path from my laptop to YouTube and the number of hops.

### 4. Checking Disconnected Device

I disconnected one device from Wi-Fi and tried pinging it again. The message showed "Request timed out", which means the device did not reply.

### 5. IP Address Check

I used the arp -a command to view the IP and MAC addresses of devices my laptop had learned about on the local network.

### 6. Results

- The laptop connected to Wi-Fi successfully.
- The ping command helped test network connectivity.
- Traceroute showed the path to YouTube.
- The disconnected device did not reply to ping.
- The ARP command displayed known network devices.

